

Low Dropout Regulator

Electronic Design Project (EDP)

by

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May 23, 2026

DECLARATION

I hereby declare that the report entitled “**Low dropout Regulator**” submitted by me, for *Electronic Design Project at IIST* is a record of bonafide work carried out by me under the supervision of **Dr. Immanuel Raja**, IIST, Thiruvananthapuram.

Place: Thiruvananthapuram

Date: 23.05.2026

Signature of the Candidate

Executive Summary

A Low Dropout Regulator (LDO) is a linear voltage regulator designed to provide a stable and noise-free output voltage even when the difference between the input and output voltage is very small. Unlike conventional linear regulators, an LDO can operate efficiently with a low dropout voltage, making it highly suitable for modern low-power electronic systems. CMOS-based LDOs are widely used in integrated circuits and portable electronic devices because of their compatibility with standard CMOS fabrication processes, low power consumption, compact area, and ease of integration.

The primary function of an LDO is to maintain a constant output voltage despite variations in input supply voltage, load current, and temperature. This regulation capability is extremely important in sensitive analog, RF, mixed-signal, and digital circuits where supply noise and voltage fluctuations can significantly affect system performance. Applications of LDOs include mobile devices, communication systems, sensor interfaces, biomedical electronics, microprocessors, and battery-powered systems.

An ideal CMOS LDO is expected to satisfy several important performance requirements. It should provide excellent line and load regulation to ensure a stable output voltage under varying operating conditions. A high Power Supply Rejection Ratio (PSRR) is essential to suppress supply noise and improve signal integrity, especially in RF and mixed-signal applications. The regulator should also exhibit low quiescent current to maximize battery life in portable systems. Fast transient response is another critical requirement, enabling the LDO to quickly recover from sudden load current changes without significant voltage overshoot or undershoot.

In addition, an ideal LDO should maintain low output voltage ripple, high current driving capability, low dropout voltage, and good thermal stability. Stability over a wide range of load capacitances and process-voltage-temperature (PVT) variations is also an important design consideration. CMOS implementation further aims to achieve reduced chip area, low fabrication cost, and improved integration with System-on-Chip (SoC) architectures.

This project focuses on the design and analysis of a CMOS Low Dropout Regulator with the objective of achieving stable voltage regulation, low power operation, and improved overall performance suitable for modern integrated electronic systems.

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CHAPTER 1

INTRODUCTION

1.1 Objective

The primary objective of this work is to design and analyze a CMOS-based Low Dropout Regulator (LDO) capable of providing a stable and reliable output voltage for low-power integrated electronic systems. The proposed design focuses on improving the overall regulation performance while maintaining low power consumption and efficient operation.

This paper mainly aims to reduce line regulation and load regulation so that the output voltage remains nearly constant despite variations in input supply voltage and load current conditions. Minimizing these parameters enhances the accuracy and stability of the regulator, which is essential for sensitive analog and mixed-signal applications.

Another major objective of this work is to improve the Power Supply Rejection Ratio (PSRR) of the LDO. A high PSRR enables the regulator to effectively suppress noise and ripple present in the supply voltage, thereby improving the performance and reliability of the connected circuitry. This is particularly important in RF, communication, and high-speed integrated systems where supply noise can degrade signal quality.

In addition to achieving low line regulation, low load regulation, and improved PSRR, the proposed CMOS LDO is also expected to provide low dropout voltage, reduced quiescent current, good transient response, and stable operation across different process and temperature variations. The overall goal is to develop an efficient, compact, and high-performance LDO suitable for modern System-on-Chip (SoC) and portable electronic applications.

CHAPTER 2

Literature Review

2.1 Review of Literature

Low Dropout Regulators (LDOs) have become an essential part of modern System-on-Chip (SoC), RF, biomedical, and portable electronic systems due to their capability of providing stable, low-noise, and regulated output voltages. Continuous advancements in CMOS technology and portable electronics have increased the demand for capacitor-less and low-power LDO architectures with improved Power Supply Rejection Ratio (PSRR), low quiescent current, fast transient response, and enhanced line/load regulation.

Chen and Hung proposed a capacitor-free CMOS LDO regulator employing enhanced active feedback (EAF) frequency compensation techniques to improve loop stability without using external load capacitors. The proposed regulator achieved stable operation for load currents up to 100 mA while using only 7 pF of compensation capacitance. Their work highlighted the importance of compensation techniques in overcoming stability issues associated with multistage amplifiers in low-voltage CMOS LDOs. The regulator also demonstrated low quiescent current and improved transient response, making it suitable for portable electronic applications.

Dossanov and Issakov presented a low quiescent current capacitor-less LDO implemented in 22 nm FDSOI CMOS technology. Their design introduced a current charge and discharge path compensation technique that improved transient response and ensured stability across wide load variations. The proposed LDO consumed only 2.5 μ A quiescent current while achieving 0.8 mV/V line regulation, 1.15 mV/mA load regulation, and a PSRR of 62 dB up to 100 Hz. Their work demonstrated that modern FDSOI technology can significantly reduce power consumption while maintaining high regulation performance for ultra-low-power biomedical and IoT applications.

Limpisawas and Wattanapanitch proposed a low-power wide-load-range output-capacitorless LDO employing indirect-direct nested Miller compensation. Their work focused on maintaining stability over a broad range of load currents and load capacitances while preserving excellent DC load and line regulation. The proposed architecture achieved load regulation of 0.025 mV/mA and line regulation of 0.5 mV/V with a quiescent current of only 14 μ A. The authors emphasized the importance of maintaining high loop gain and effective compensation techniques to ensure stability and high-performance operation in modern SoC environments.

Li et al. introduced a cascoded flipped voltage follower (CAFVF) based output-capacitorless LDO regulator for SoC applications. By incorporating a cascode current source and cascode compensation techniques, the proposed architecture significantly improved loop gain and bandwidth while reducing minimum load current requirements. The design achieved 58.6 dB DC gain under heavy load conditions and extended unity gain frequency from 1.51 MHz to 2.36 MHz. Additionally, the regulator operated with an ultra-low quiescent current of 14 μ A and a dropout voltage of 200 mV. Their work highlighted the effectiveness of CAFVF architectures in enhancing load regulation and transient performance.

Park and Silva-Martinez investigated design techniques for external capacitor-less LDOs with high PSRR over a wide frequency range. Their research analyzed different power supply noise leakage paths in LDO regulators and proposed adaptive supply noise compensation techniques to minimize noise

coupling to the output. The developed LDO achieved PSRR better than -50 dB up to 4 MHz while operating without bulky external capacitors. Their work established the significance of PSRR enhancement techniques in mixed-signal and RF integrated circuits where suppression of switching noise is critical.

Hanumolu provided a comprehensive mathematical and conceptual overview of LDO regulators, discussing important performance parameters such as dropout voltage, quiescent current, efficiency, line regulation, load regulation, transient response, and PSRR. The work explained the role of feedback mechanisms, pass devices, and output capacitors in maintaining output voltage stability under varying supply and load conditions. The study also emphasized that high-performance LDOs require careful trade-offs among stability, noise suppression, power efficiency, and transient behavior.

2.2 Conclusions from the literature review

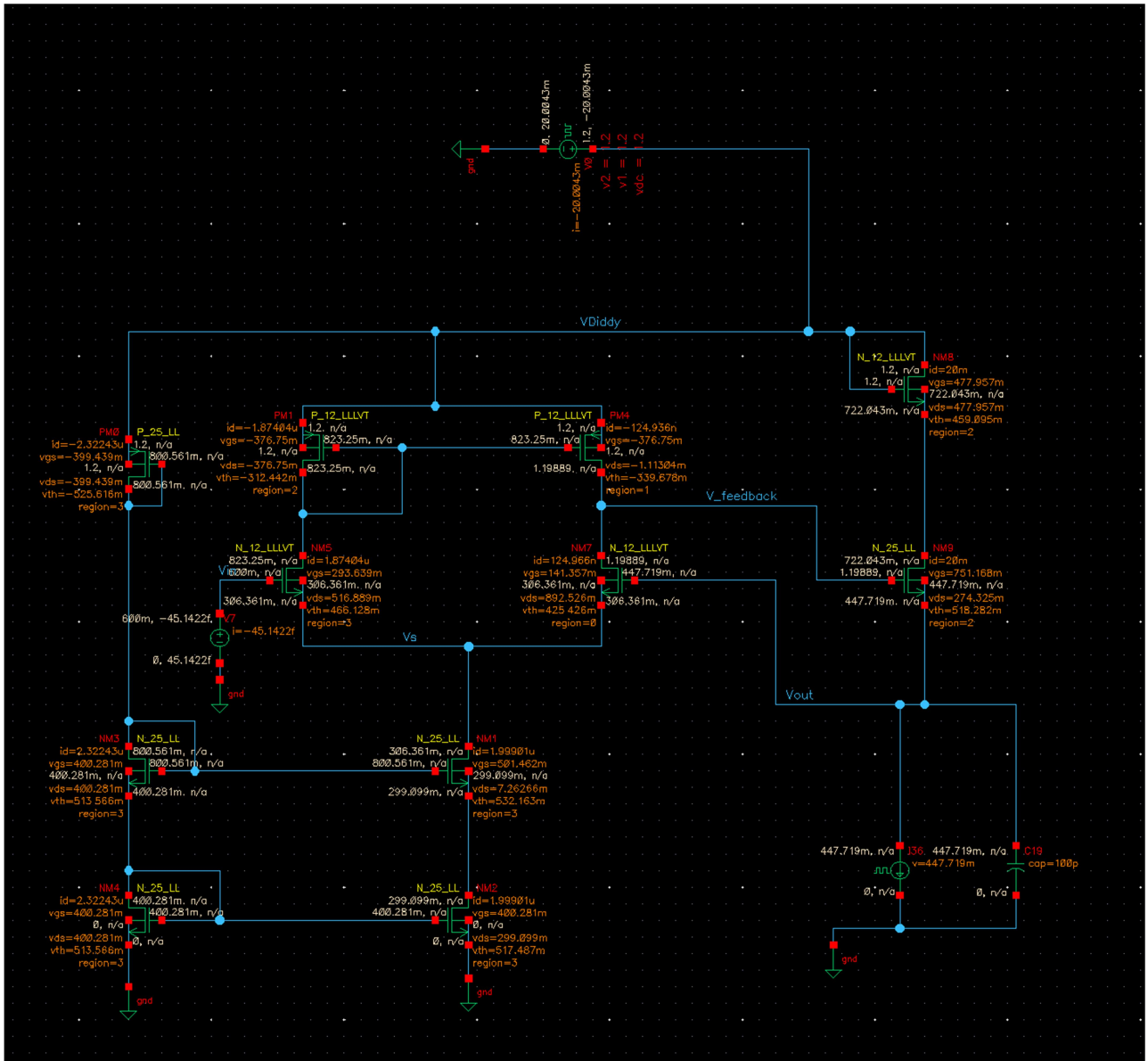
From the reviewed literature, it is observed that recent research trends mainly focus on improving stability, reducing quiescent current, enhancing PSRR, and achieving better line and load regulation in capacitor-less CMOS LDO architectures. Different compensation methods such as enhanced active feedback, nested Miller compensation, cascode compensation, and adaptive noise cancellation have been explored to achieve stable operation over wide load conditions. Based on these studies, the present work aims to design a CMOS LDO regulator with improved PSRR, reduced line and load regulation, low power consumption, and stable operation suitable for modern low-power integrated systems.

CHAPTER 3

Architecture

In this project we have mainly developed 3 LDO architectures with different characteristics to cover wide range of applications.

Architecture 1:



The proposed circuit is a CMOS-based Low Dropout Regulator (LDO) designed to provide a stable regulated output voltage with low power consumption and low dropout operation. The circuit mainly consists of a differential error amplifier, current mirror biasing circuitry, a PMOS pass transistor, a feedback network, and an output compensation capacitor. The regulator operates using a closed-loop negative feedback mechanism to maintain voltage regulation under varying input supply and load conditions.

In this circuit, the differential error amplifier continuously compares the reference voltage with the feedback voltage obtained from the output node. The difference between these two signals generates an error voltage that controls the gate terminal of the NMOS pass transistor. Based on the generated error signal, the conduction of the pass transistor is dynamically adjusted to maintain a constant output voltage.

When the output voltage decreases due to an increase in load current, the feedback voltage also decreases. This causes the pass transistor to conduct more current, thereby restoring the output voltage to the desired level. Similarly, when the output voltage increases, the conduction of the pass transistor is reduced, resulting in stabilization of the output voltage.

In this circuit we have used a mix of N_12_LLVT and N_25_LLVT technology node because of the following advantages of N_25_LLVT:

- Better voltage handling capability
- Improved reliability
- Lower leakage compared to thin oxide devices
- Better robustness

Circuit Block Analysis

NMOS Pass Transistor

In this circuit, the NMOS transistor connected near the output stage acts as the pass element of the regulator. The NMOS pass transistor controls the amount of current supplied to the load based on the error signal generated by the amplifier stage.

The use of a NMOS pass transistor offers several advantages as given in the table.

Parameter	PMOS Pass Transistor	NMOS Pass Transistor
Mobility	Lower	Higher
Area	Larger	Smaller
Current Driving Capability	Moderate	High
Transient Response	Slower	Faster
Gate Drive Complexity	Simple	Complex
PSRR	Moderate	Better at high frequency
Stability	Easier	More challenging

NM8: Acts as a current source used to supply current to the pass transistor when current load increases helps to improve load regulation.

NM9: Used as a pass transistor so that if V_{out} decreases it helps us to control the decreasing V_{out} due to current load.

NM2 and NM1: used as a cascode current mirror stage to supply high current in the circuit.

PM0: Acts as a diode connected current source

The large dimension of the NMOS transistor improves the current driving capability and reduces the effective ON resistance, thereby improving load regulation performance. However, the large gate capacitance introduced by the NMOS device also contributes to stability challenges within the feedback loop.

Differential Error Amplifier

In this circuit, the differential amplifier section is formed using PMOS and NMOS transistors in a 5 transistor OTA configuration acting as an active differential pair with current mirror loading. The amplifier compares the reference voltage with the feedback voltage derived from the output.

The differential amplifier provides voltage gain and improves the sensitivity of the regulator to variations in output voltage. A higher loop gain enhances both line regulation and load regulation by minimizing output voltage deviations caused by variations in supply voltage and load current.

The amplifier stage also plays an important role in determining the transient response and Power Supply Rejection Ratio (PSRR) of the regulator.

Current Mirror Biasing Circuit

The lower section of the circuit consists of NMOS current mirror transistors that establish stable bias currents for the amplifier stages. The current mirror network ensures proper operating conditions for all transistors and contributes to low quiescent current operation.

In this circuit, the bias currents are maintained at relatively small values to reduce overall power dissipation. This makes the design suitable for low-power and battery-operated applications.

However, current mirror mismatch and process variations may slightly affect the accuracy and stability of the regulator performance.

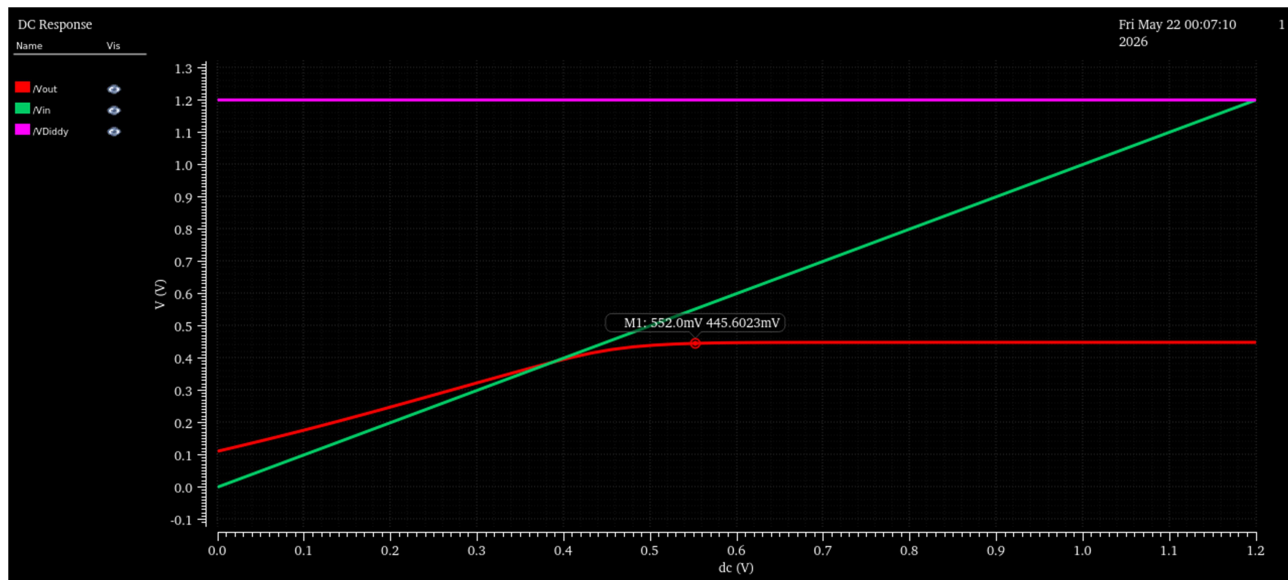
Output Capacitor

A small output capacitor of approximately 100 pF is connected at the output node to improve loop stability and reduce output voltage ripple.

The use of a small capacitor makes the circuit more suitable for System-on-Chip (SoC) applications where large external capacitors are undesirable. The output capacitor also improves transient response by supplying instantaneous current during sudden load variations.

However, the reduced output capacitance increases the difficulty of maintaining stability over a wide range of load conditions.

Performance Analysis



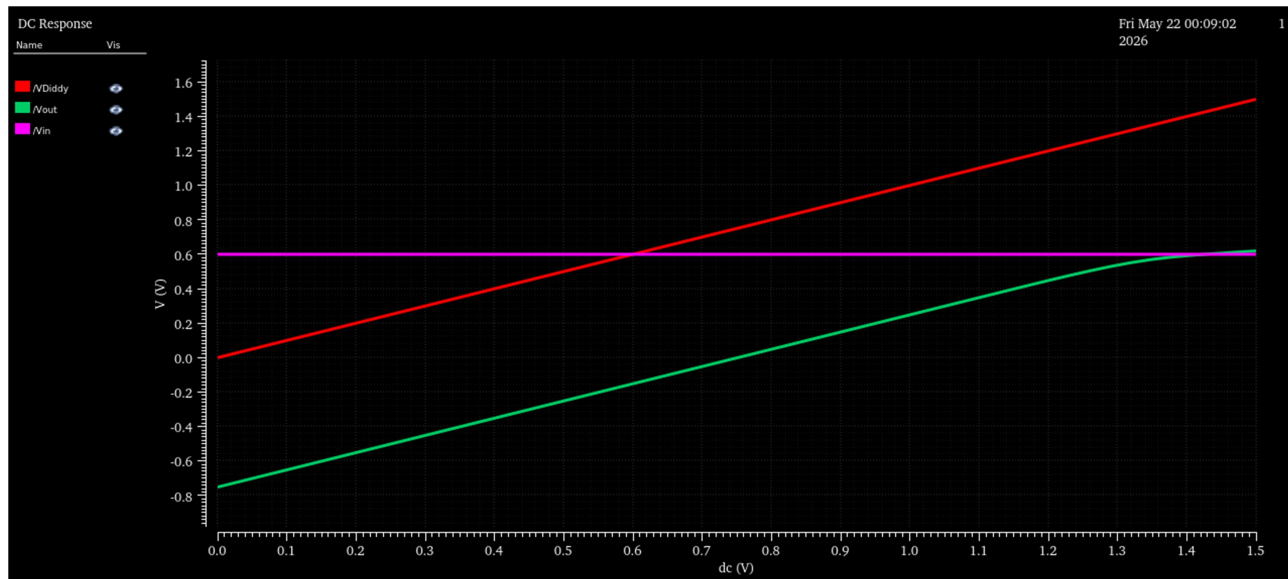
In DC Analysis, we sweep V_{in} to observe how V_{out} follows the sweeping V_{in} . The circuit successfully enters regulation mode right from $V_{in} = 0$ V.

Linear: V_{out} is linear with V_{in} . After $V_{in} = 552$ mV, output goes constant.

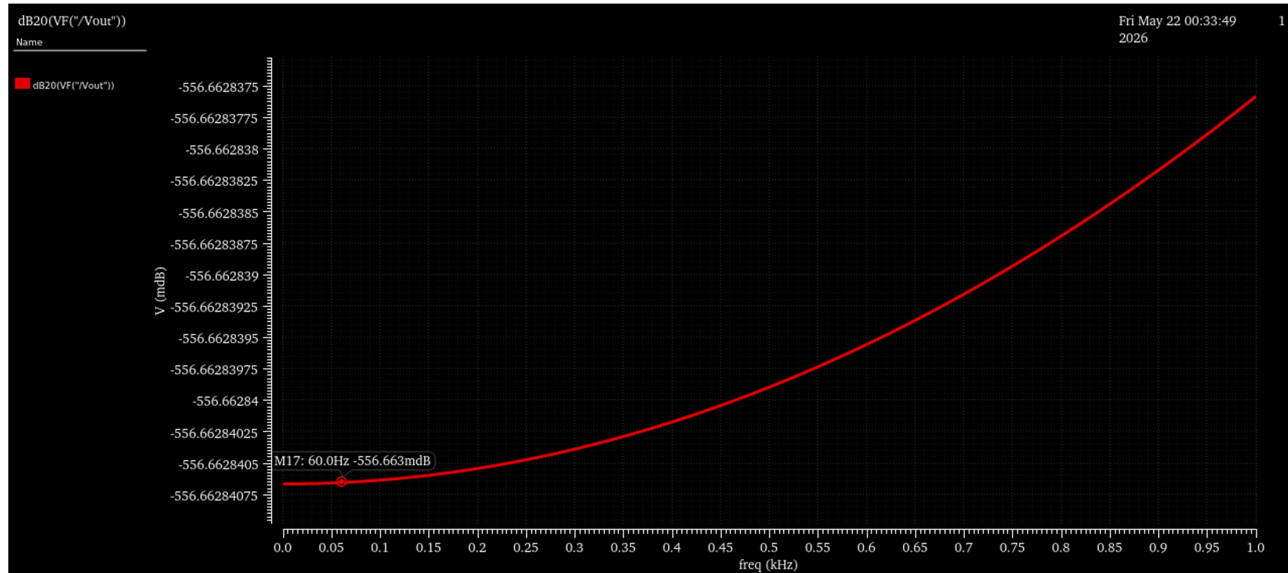
Good Line Regulation: V_{out} follows V_{in} from 0 to 552 mV.

Conclusion: This circuit can regulate output voltage from 0 to 552 mV.

Low offset Output voltage of 0.1 V is observed.

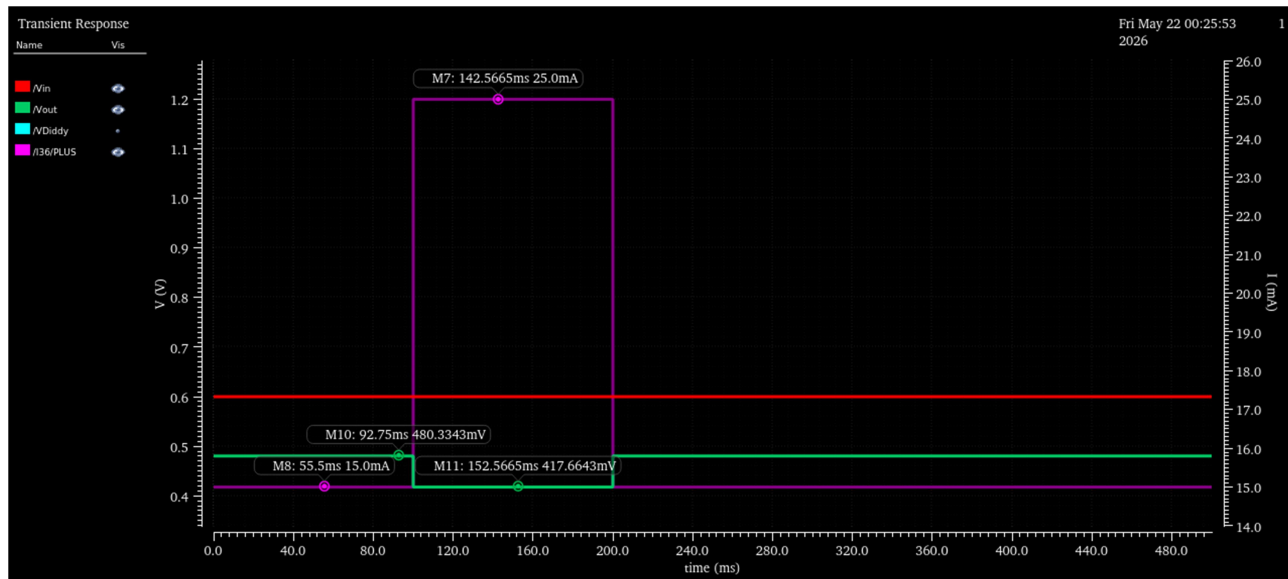


In this DC Analysis, we have kept a constant output load of 10mA and have fixed $V_{in} = 0.6V$. Here we are measuring the variation of V_{out} with sweeping VDD. Here we can clearly see that V_{out} saturates at V_{in} voltage which is what we expect from our LDO.



This is an AC analysis where we supplied 1V AC at VDD to measure PSRR (Power supply rejection ratio). It measures the circuits ability to suppress voltage ripples at power supply appearing at the output.

Here PSRR is measured to be -0.5m db.

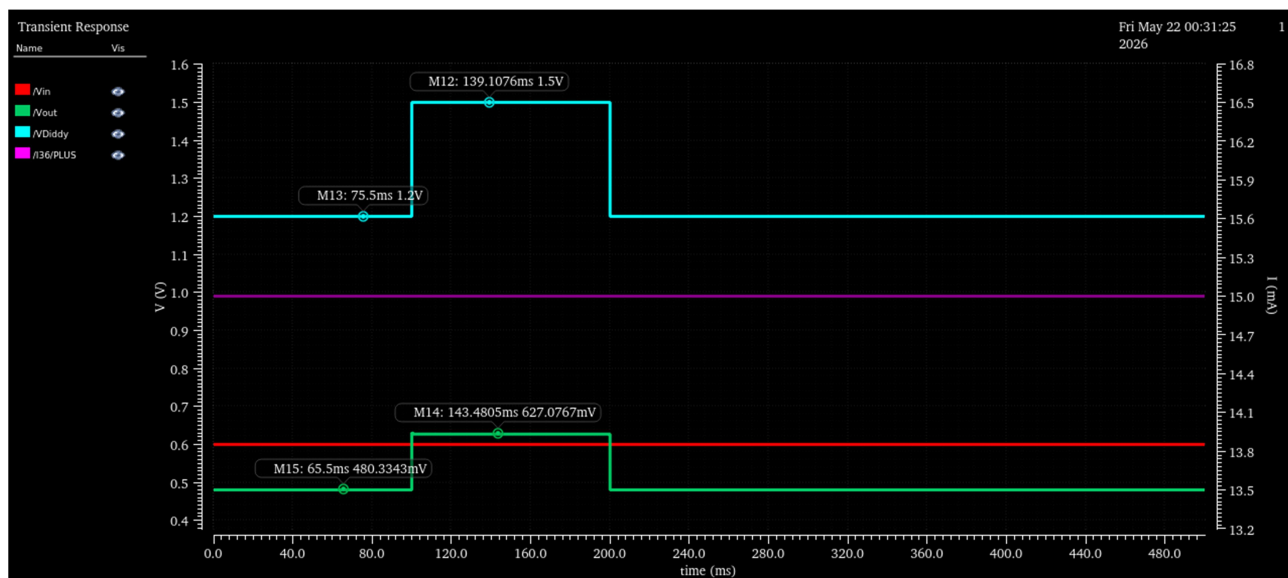


Load regulation: Change in Vout/Change in I_{load}

Change in Vout = 480mV-417mV = 63mV

Change in I_{load} = 25mA-15mA = 10mA

Load regulation = 6.3 V/A



Line regulation = Change in Vout/Change in VDD

Change in Vout = 147mV

Change in VDD = $1.5 - 1.2 = 0.3$

Line regulation = $0.147/0.3 = 0.5 \text{ V/V}$

Advantages of the Proposed Circuit

- Simple CMOS architecture suitable for integrated circuit implementation.
- Small output capacitor requirement enables easier SoC integration.
- Good load current driving capability upto 25mA.
- Reduced external component requirement.
- Compact and low-power design.

Conclusion

The proposed CMOS LDO regulator provides an efficient low-power voltage regulation solution with low dropout operation and compact implementation. In this circuit, the NMOS pass transistor, differential amplifier, and current mirror biasing network collectively maintain stable output voltage under varying input and load conditions. The use of a small output capacitor makes the design suitable for modern System-on-Chip applications.

The circuit offers advantages such as low quiescent current, compact area, and good regulation capability.

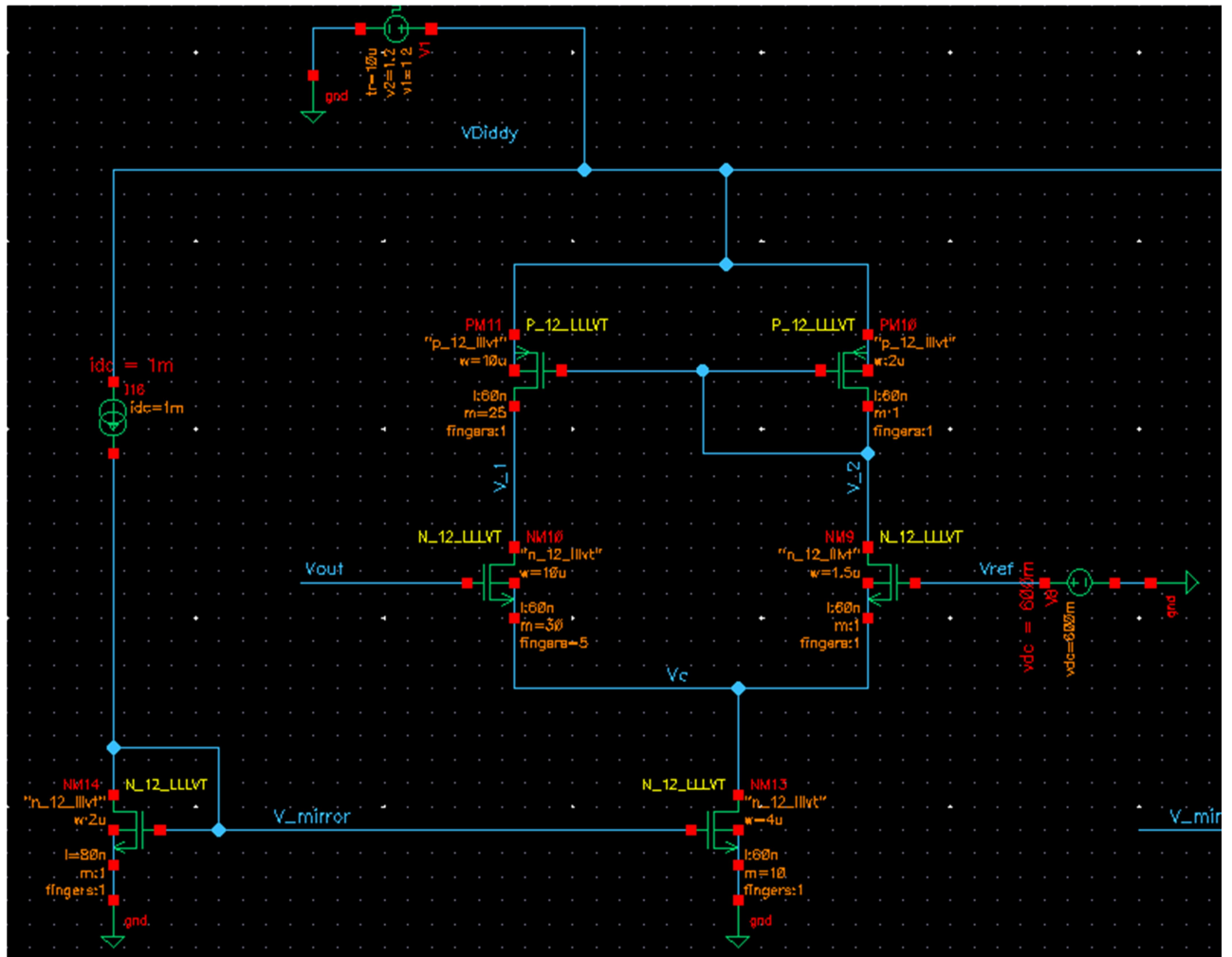
PSRR: -0.5 db

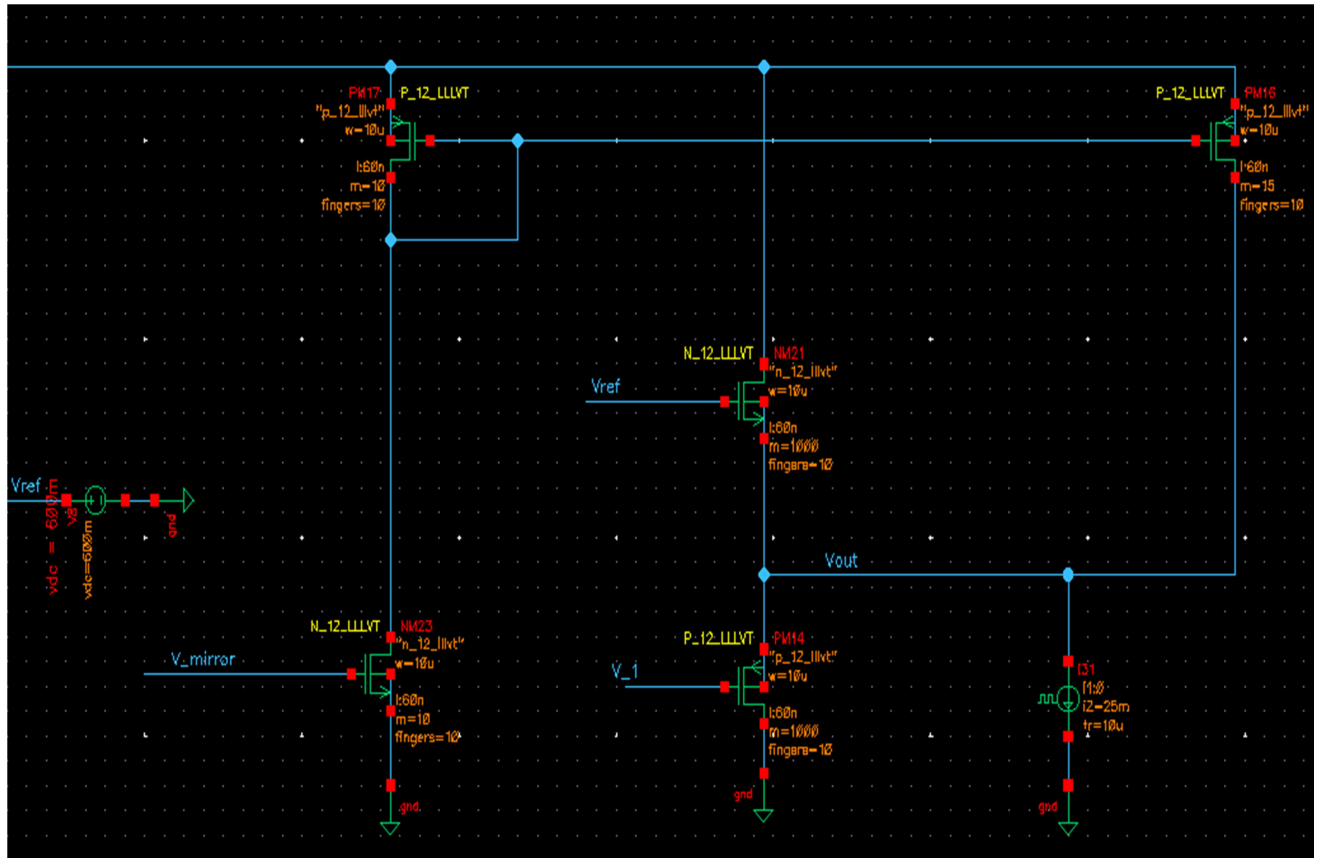
Line Regulation: 0.5 V/V

Load Regulation: 6.3 V/A

V_{in} Range: 0-552mV

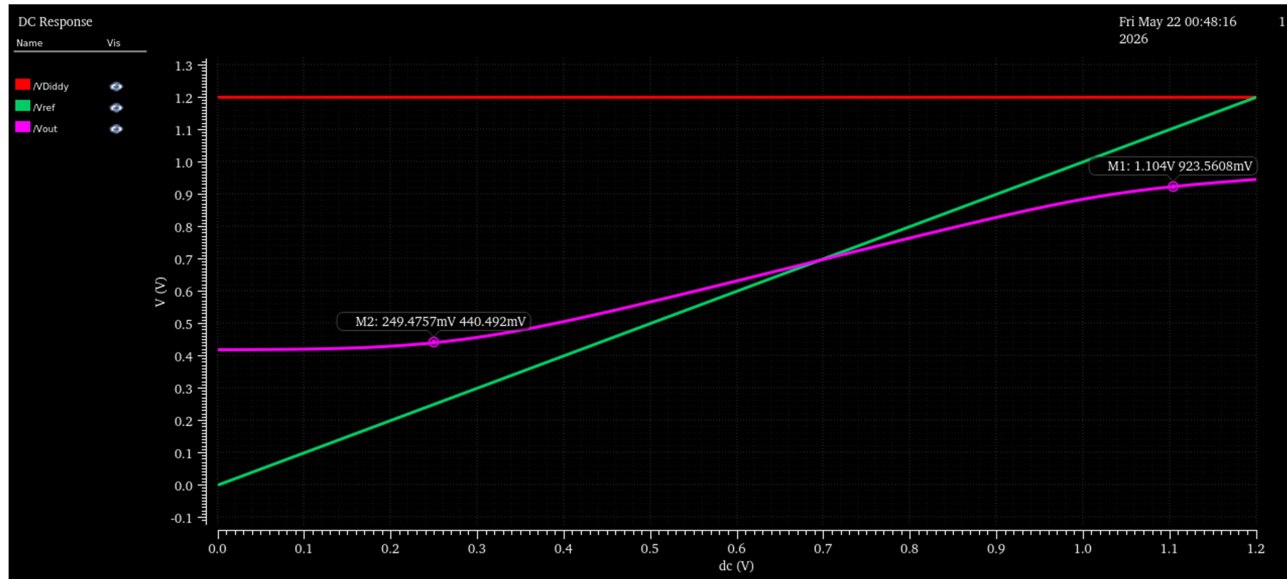
Architecture 2:



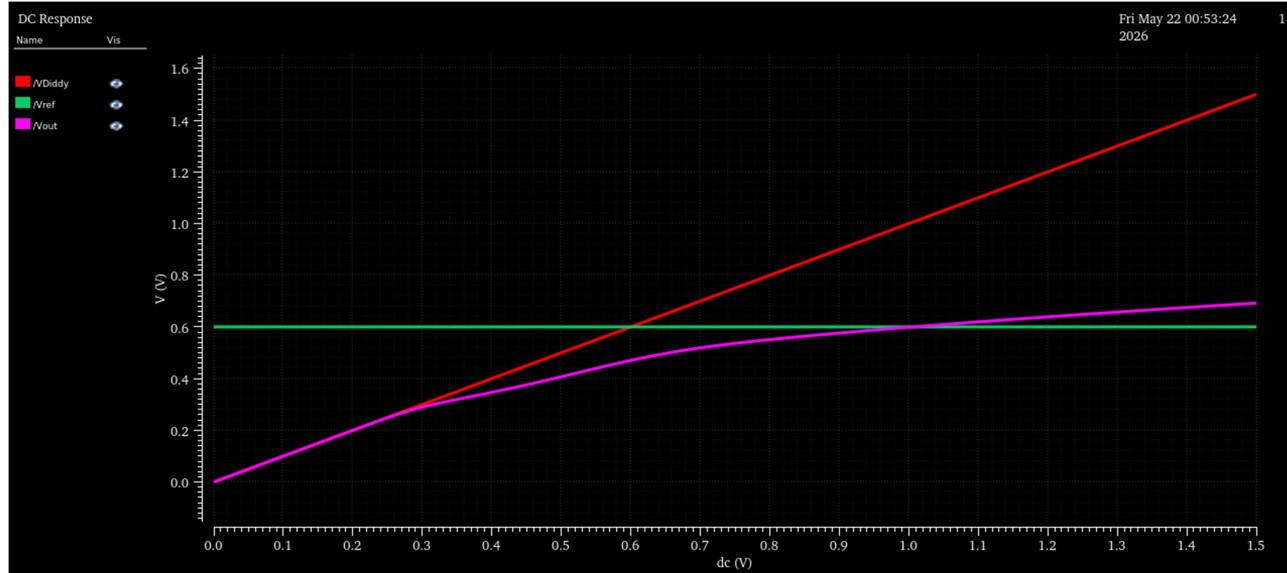


In this topology, both NM21 and PM14 is acting as a pass transistor which maintains the output voltage by comparing V_{ref} with V_{out} . If $V_{out} < V_{ref}$ then NM21 allows current from power supply to increase V_{out} . If $V_{out} > V_{ref}$ then PM14 allows ground potential to decrease V_{out} .

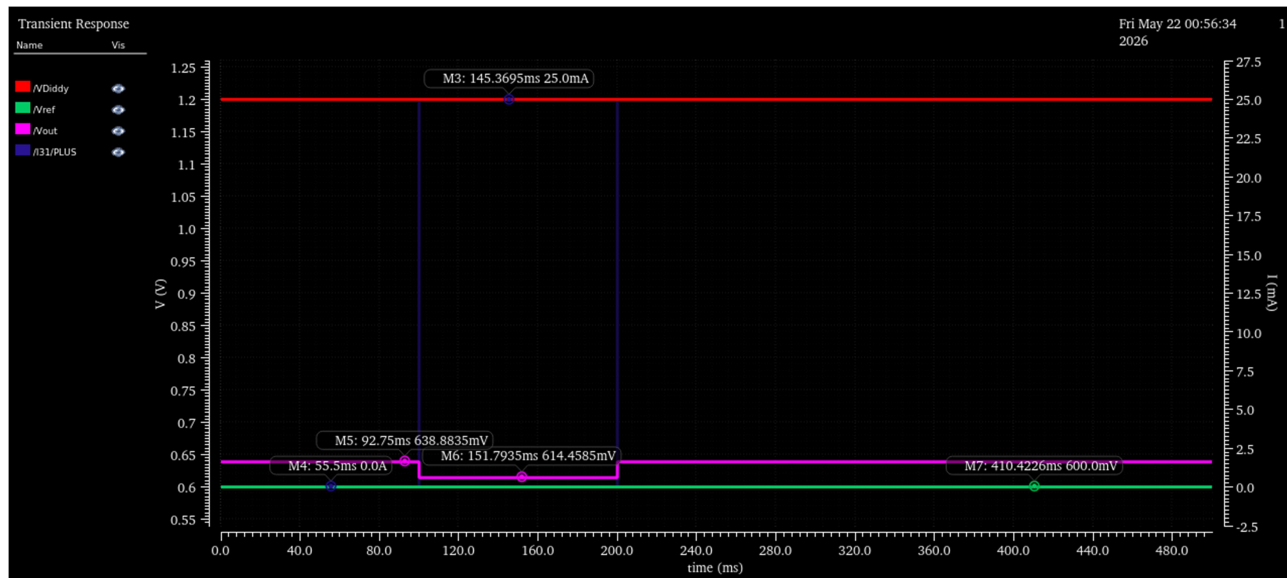
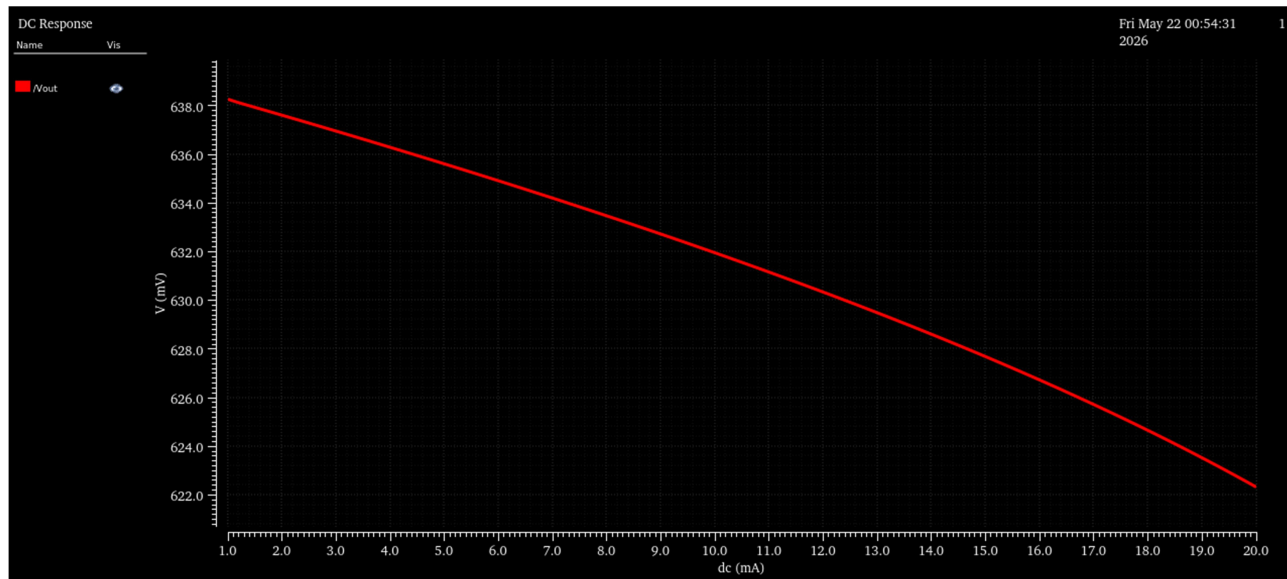
PM16 is acting as a current source to supply heavy demand of I_{load} at the output.



The graph is the DC Sweep of V_{ref} from 0 to 1.2V. We expect V_{out} to vary linearly with V_{in} . Here we can clearly see that V_{out} responds to the changing V_{ref} after 250mV and follows till 1 V. But here we can see a large offset of 249mV at the output due to PM16, which is a potential drawback of this topology.



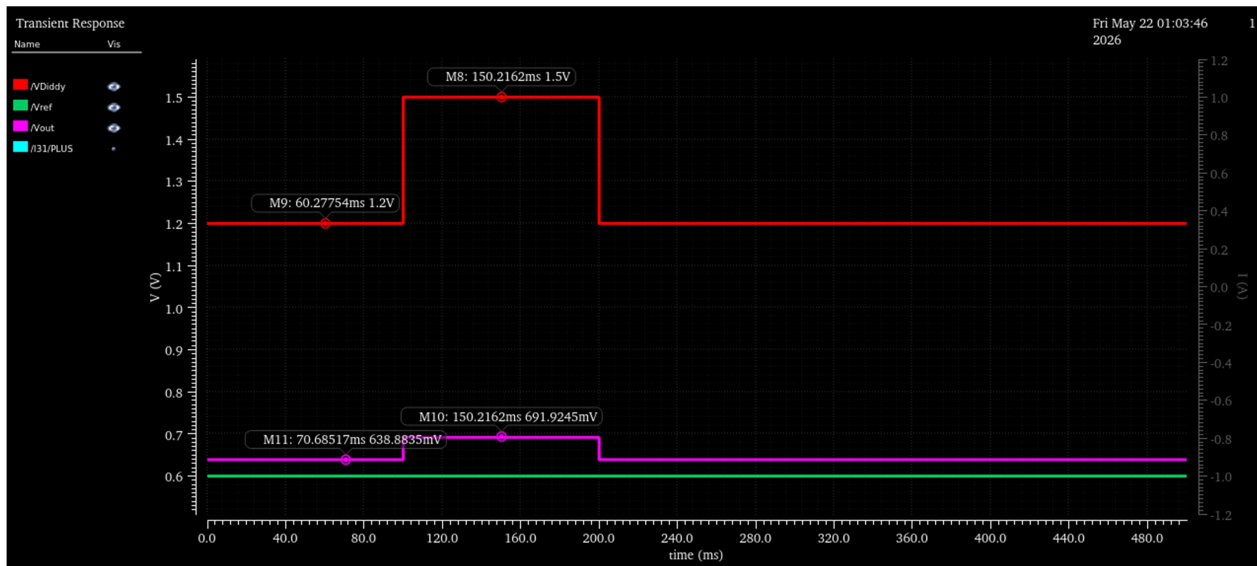
This graph is DC Sweep of V_{DD} . As expected, V_{out} saturates at V_{ref} which shows V_{ref} has good control at the output node.



Both these graphs shows us drop in V_{out} due to increasing I_{load} current.

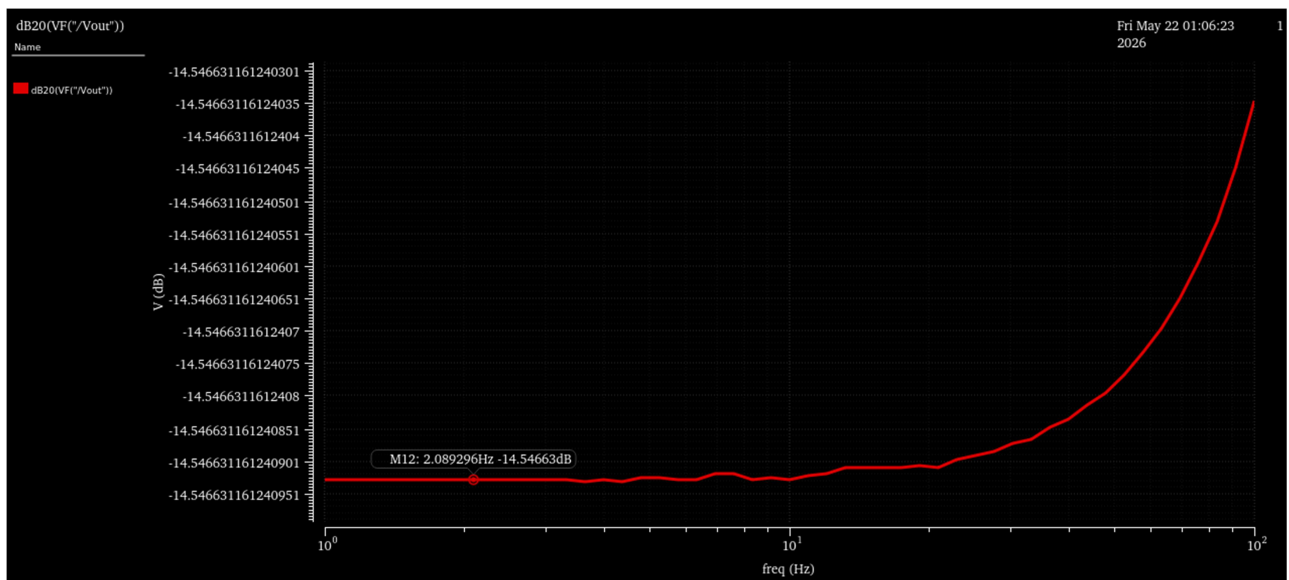
Load Regulation: $\text{Change in } V_{out} / \text{Change in } I_{out}$

Load Regulation = $638 - 614 \text{ mV} / 20 \text{ mA} = 1.2 \text{ V/A}$



Line Regulation = Change V_{out} / Change VDD

Line Regulation = $691 - 638 \text{ mV} / 1.5 - 1.2 = 0.17 \text{ V/V}$



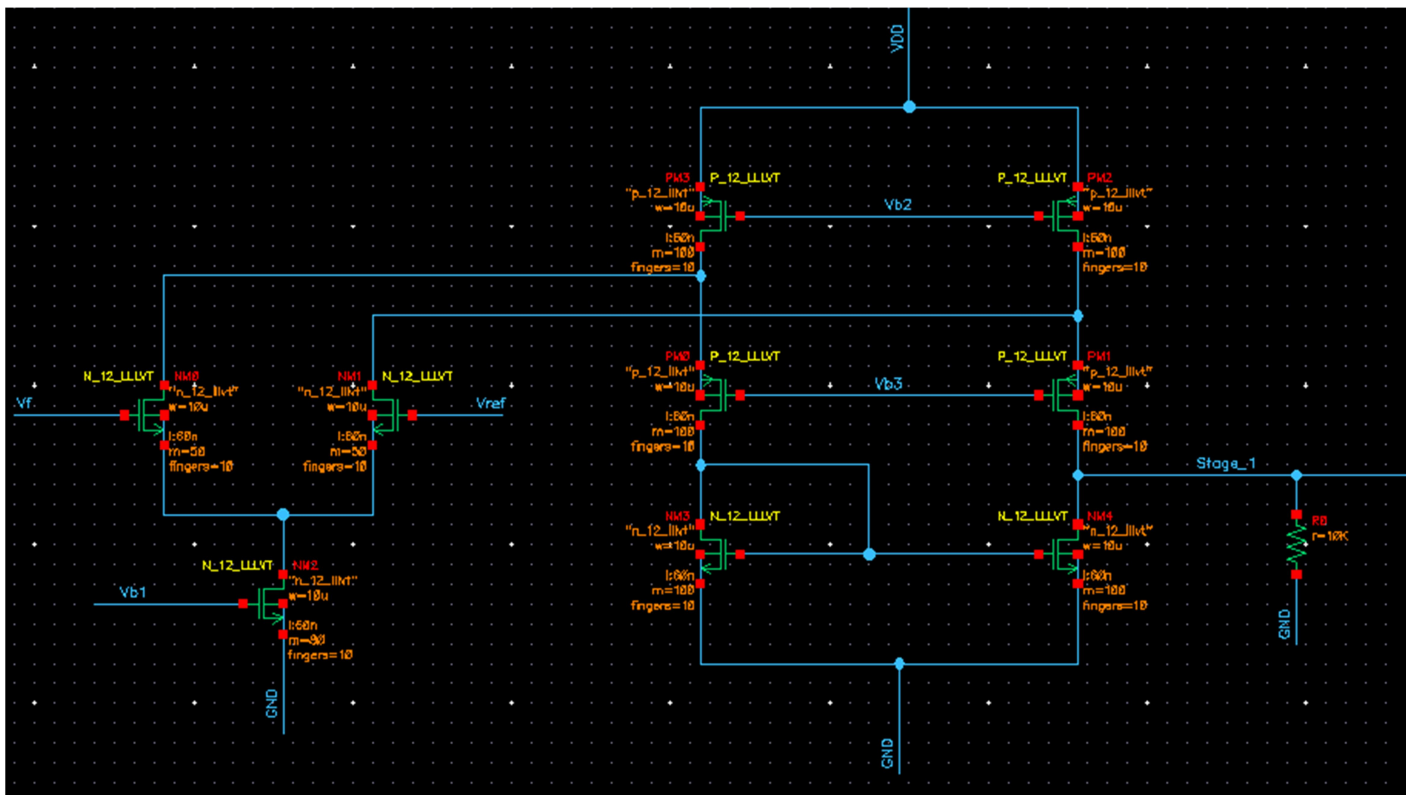
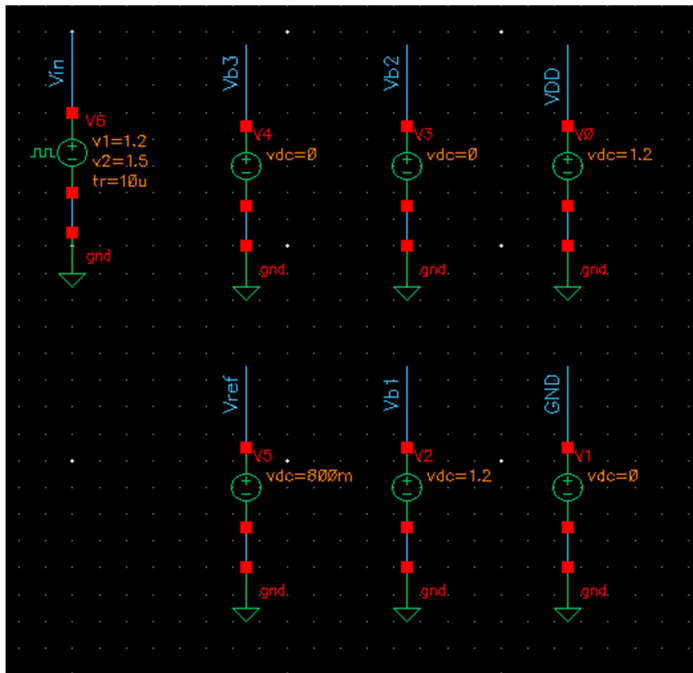
PSRR = -14 dB

Line Regulation = 0.17 V/V

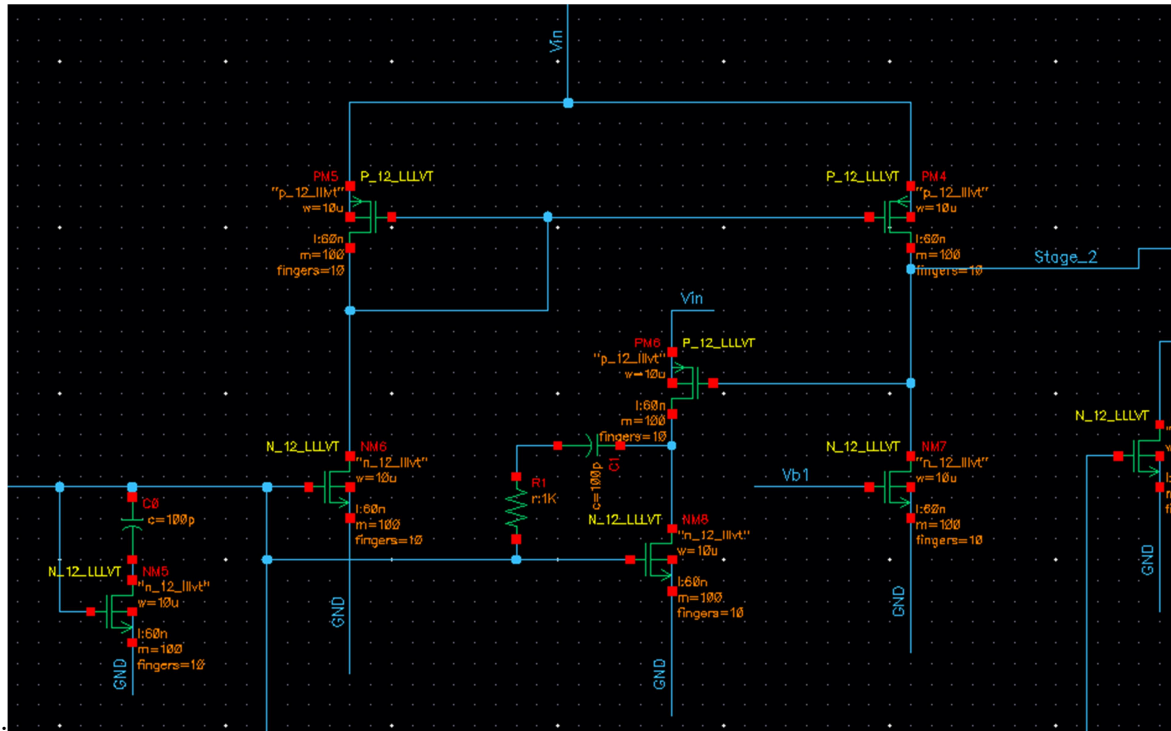
Load Regulation = 1.2 V/A

Vref range: [250mV, 1V]

Architecture 3:

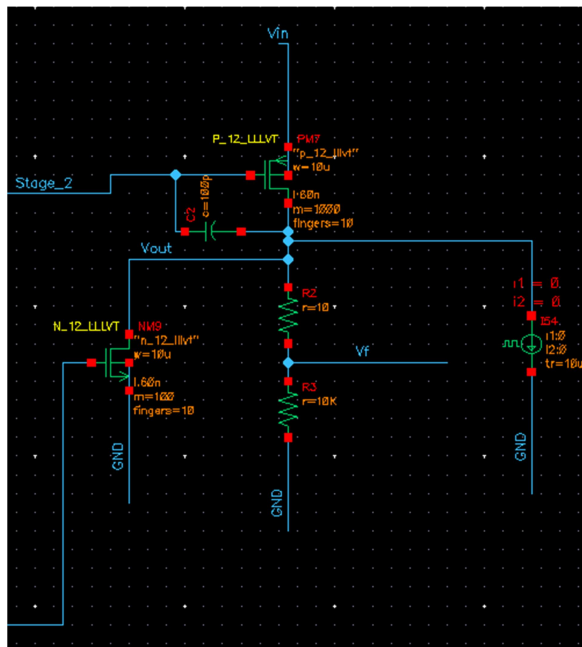


In this topology we have used telescopic cascode for high gain.

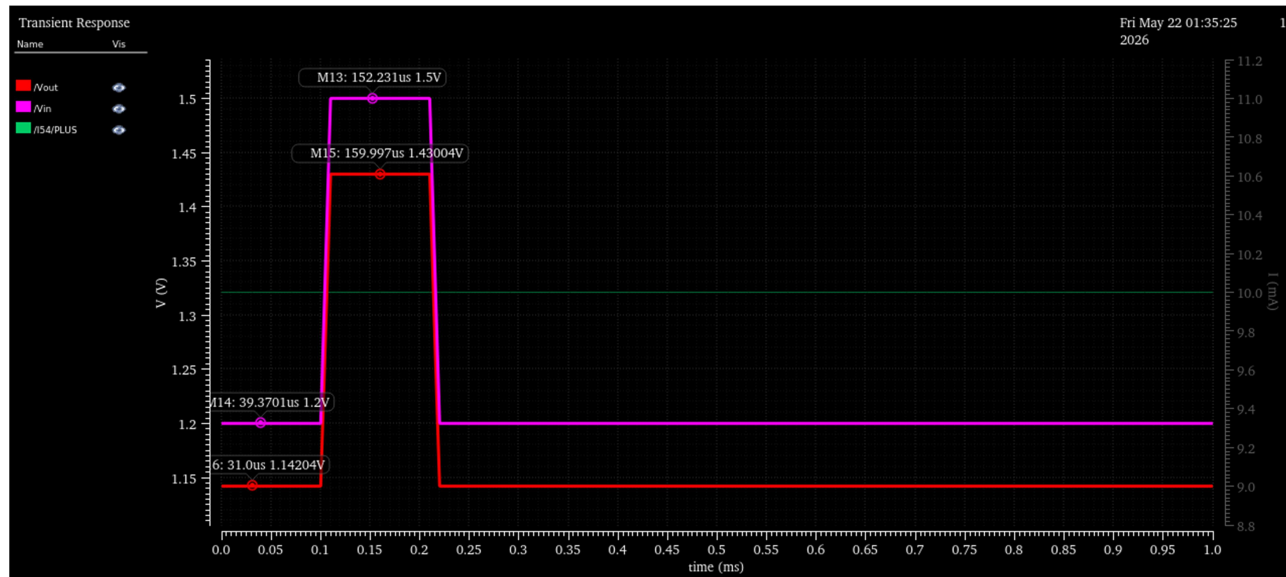


In this circuit, NM6 and NM7 form the NMOS differential input pair. PM5 and PM4 act as active current mirror loads. The differential pair compares feedback voltage, reference voltage and converts the voltage difference into differential current.

The PMOS current mirror load converts this differential current into a single-ended voltage output at the node labelled as Stage_2. This node drives the next amplification stage or the gate of the pass transistor.

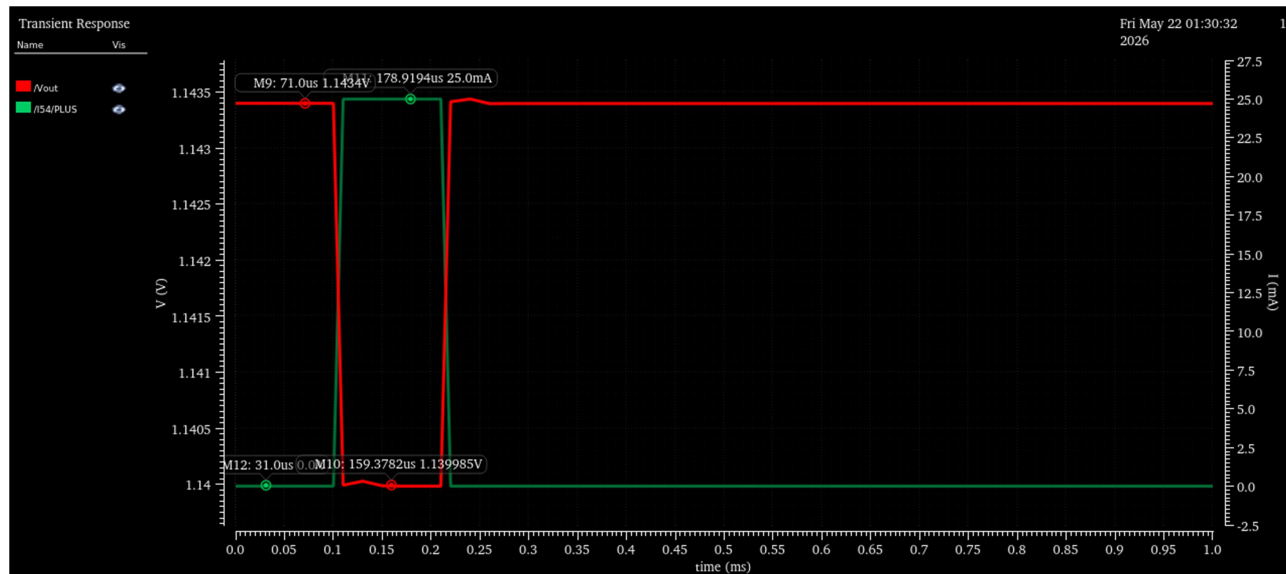


The capacitor C2 acting as a miller capacitor improves loop stability, phase margin, transient response as well as it also introduces dominant pole compensation, pole splitting effect.



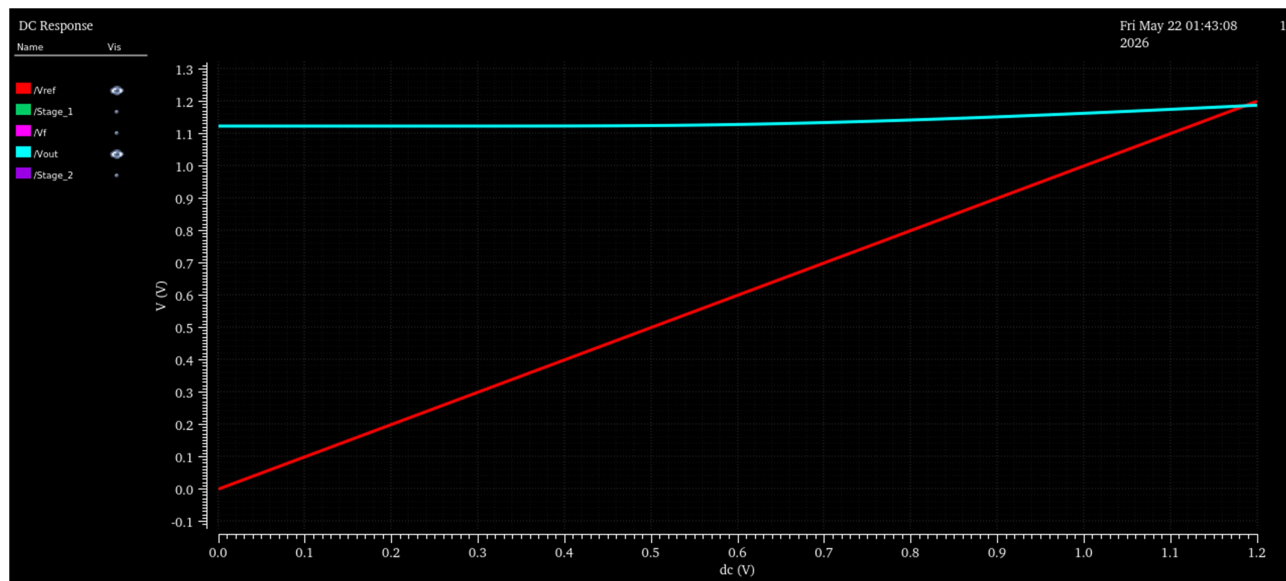
Line Regulation: Change in V_{out} / Change in VDD

Line regulation = $1.43 - 1.142 / 1.5 - 1.2 = 0.288 / 0.3 = 0.96 \text{ V/V}$

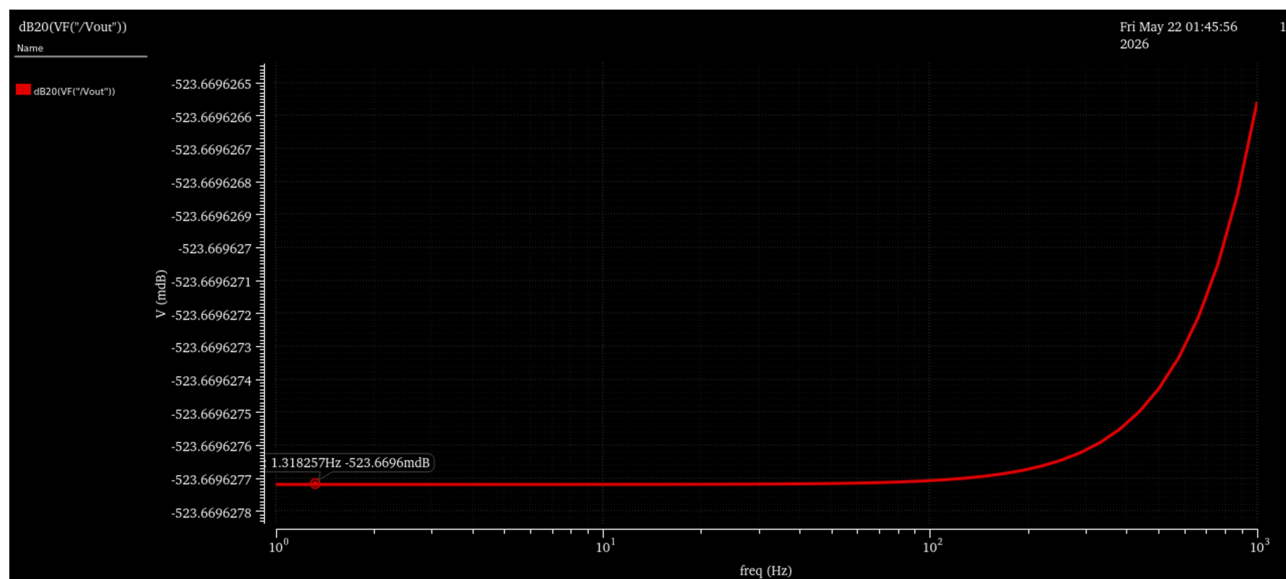


Load Regulation = Change in V_{out} / Change in I_{load}

Load Regulation = $1.143 - 1.14 \text{ V} / 25\text{mA} = 0.12 \text{ V/A}$



In this architecture, V_{out} is virtually independent of V_{ref} . Thus, its range of operation is unrecognizable.



PSRR = -0.5 dB

Line regulation = $1.43 - 1.142 / 1.5 - 1.2 = 0.288 / 0.3 = 0.96 \text{ V/V}$

Load Regulation = $1.143 - 1.14 \text{ V} / 25\text{mA} = 0.12 \text{ V/A}$

CHAPTER 4

RESULTS AND DISCUSSION

Feature	Architecture 1	Architecture 2	Architecture 3
Range	0-552mV	250mV – 1V	
Line Regulation	0.5 V/V	0.17 V/V	0.96 V/V
Load Regulation	6.3 V/A	1.2 V/A	0.12 V/A
PSRR	-0.5 dB	-14 dB	-0.5 dB
Linearity	Very high	moderate	poor
Offset voltage	0.1 V	0.4V	1.2 V

Architecture 1 operates only up to 552 mV, indicating that it is suitable for ultra-low-voltage applications. However, the operating range is comparatively limited. Architecture 2 provides a wider operating range from 250 mV to 1 V, demonstrating better adaptability and improved regulation capability over a broader supply range. This makes Architecture 2 more practical for low-power integrated systems. The operating range for Architecture 3 is virtually unrecognizable.

Line regulation represents the ability of the LDO to maintain constant output voltage despite variations in input supply voltage. Lower values indicate better performance. Among the three architectures, architecture 2 exhibits the best line regulation with a value of 0.17 V/V. This indicates that Architecture 2 is most effective in suppressing output voltage variations caused by changes in input voltage. Architecture 1 shows moderate performance with 0.5 V/V. Whereas, Architecture 3 exhibits the worst line regulation 0.96 V/V indicating poor immunity to supply voltage variations.

Load regulation indicates the ability of the LDO to maintain constant output voltage under varying load current conditions. Smaller values represent better regulation capability. From the table architecture 3 provides the best load regulation 0.12 V/A. This indicates excellent output voltage stability during load current variations. Architecture 2 provides moderate performance 1.2 V/A, whereas architecture 1 shows the poorest load regulation 6.3 V/A which indicates significant output voltage variation under load changes.

Power Supply Rejection Ratio (PSRR) indicates the capability of the LDO to suppress supply noise and ripple. More negative PSRR values indicate better noise rejection. Among the architectures architecture 2 demonstrates significantly better PSRR –14 dB. This indicates improved rejection of supply noise and ripple. Architectures 1 and 3 exhibit poor PSRR performance –0.5 dB which suggests limited capability in suppressing supply noise. Thus, Architecture 2 is more suitable for noise-sensitive applications such as RF and mixed-signal systems.

Linearity indicates how accurately the output voltage follows the expected regulation behavior over varying operating conditions. Architecture 1 exhibits very high linearity, indicating highly predictable and

stable voltage regulation characteristics. Architecture 2 provides moderate linearity, representing acceptable but less ideal behaviour. Architecture 3 shows poor linearity, suggesting larger deviations from ideal regulation characteristics. Thus, Architecture 1 provides the best linear operating behaviour among the three architectures.

Offset voltage represents the deviation between the desired and actual regulated voltage due to circuit imperfections. Lower offset voltage indicates better accuracy. Architecture 1 exhibits the lowest offset voltage 0.1 V indicating the highest regulation accuracy. Architecture 2 shows moderate offset voltage. Architecture 3 exhibits the largest offset voltage 1.2 V which indicates poor output accuracy and larger deviation from the intended regulated voltage. Therefore, Architecture 1 provides the most accurate voltage regulation.

CHAPTER 5

SUMMARY

The comparison indicates that each architecture is optimized for different performance objectives.

Architecture 1 demonstrates excellent linearity and the lowest offset voltage, indicating superior voltage accuracy and stable regulation behavior. However, its poor load regulation and limited operating range reduce its effectiveness under dynamic loading conditions.

Architecture 2 provides the best overall balance among the three architectures. It achieves superior line regulation, wider operating range, and significantly improved PSRR performance. These characteristics make Architecture 2 more suitable for practical low-power and noise-sensitive integrated applications.

Architecture 3 demonstrates outstanding load regulation capability, indicating excellent current driving performance and output stability under varying load conditions. However, its poor line regulation, poor linearity, and large offset voltage significantly reduce overall accuracy and regulation quality.

Overall, Architecture 2 can be considered the most balanced and practically efficient architecture due to its improved regulation capability, wider operating range, and superior noise rejection performance.

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